

Login and Registration

This task provides you with experience in [Firebase](#) and React routes. You are given the requirements to design and implement a login & registration page for your DEV@Deakin web platform.

Complete the following:

1. Complete the Task Requirements.
2. You will find “Mini-Lecture and Demo Videos” of Week 6 and Week 7 on the SIT313 CloudDeakin site particularly useful as a reference for this task.
3. Create a new [GitLab](#) repository for this task. Ensure all files/resources created are committed to this repository.
4. Create a 2 – 4 minute video where you provide a walkthrough of your code and solution, demonstrating your understanding of the concepts. Upload your video to [Panopto](#).
5. Create a short report that includes:
 - a. Screenshots of your DEV@Deakin web platform.
 - b. A link to your task repository on GitLab.
 - c. A link to your video.
6. **Upload** to OnTrack and **engage with feedback** to get this complete!

Task Requirements

1. GitLab repository

- Create a New Project called Task P4. Set the Visibility Level to **Private**.
- Once the project is created, under **Manage->Members** add the Unit Chair (Aaron Spence) and your OnTrack marking tutor **as a Reporter role**.

- Strictly do not submit your `node_modules` folder to your repository. To prevent this, create a `.gitignore` file, and add inside it `node_modules/` to prevent this folder from being tracked by Git.
- Commit all code and resources (e.g. images) that you develop for this task into this repository, with proper commit comments.

Failure to follow the above steps means your task cannot be marked as Complete.

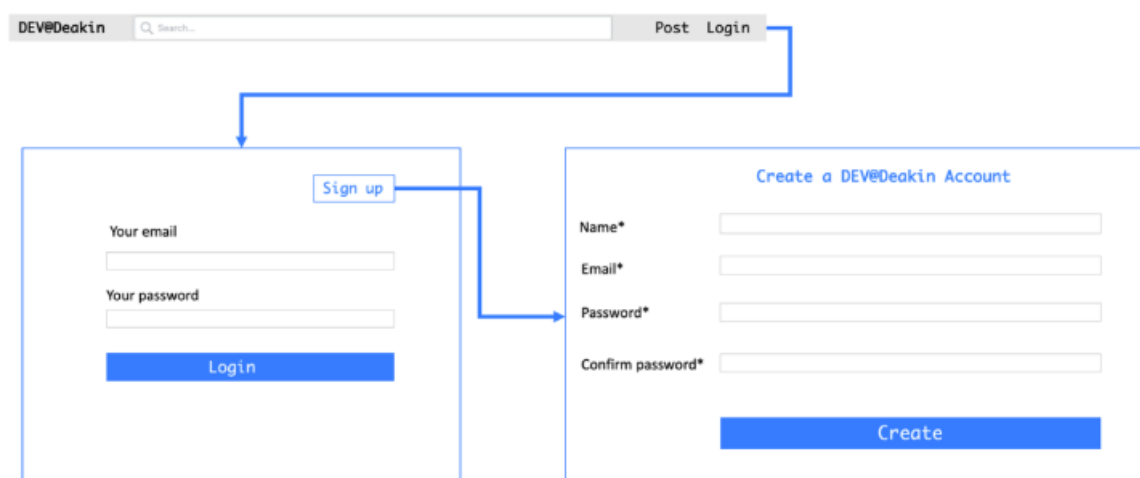
2. Navigation Bar

Modify your existing navigation bar to match exactly the below design. The DEV@Deakin link should link back to your home page when pressed. We will implement the Search and Post features in a future task.



3. Design Criteria

Review then implement into your DEV@Deakin web platform the following design criteria – your solution must match *exactly* the same formatting. However, you are free to incorporate your own designs by choosing your own styles (e.g. colours, shading, fonts, etc) and adding additional features/components/media as you wish.



4. Login Page and Sign-Up Page

The Login page allows a user to login to their existing account, or redirect to the Sign-up page to create a new account. Your solution should utilise a [Firestore](#) (a database hosted on [Firebase](#)) to maintain a record of current users.

Creating a Firebase account is free, and there are lots of guides available in the SIT313 CloudDeakin page or online.

Caution: API keys Best Practices

Never store API keys and configurations directly within your code. Best practice is to store them in an `.env` file. Also add `.env` to your `.gitignore` file.

When a user attempts to login you must check their email and password against the Firestore database:

- If their details are correct, redirect them automatically to the home page.
- If their details are incorrect, suggest the user tries again, or that they sign-up for a free account.

When a user signs-up for a new account, ask for their full name (first and last name), email address, and password.

You must conduct proper data validation before proceeding (did they input a valid details?). Consider the **Optional Technologies** section to assist here. When the details are valid, store them into your Firestore database, then redirect them automatically to the login page.

You should never store sensitive information like passwords as plain-text. However, it is not ideal to encrypt passwords via the frontend due to performance issues and interception threats, so we will allow it for now. Ideally it should be handled by the backend, which we will do in a future task.

5. Navigation Bar and Footer

Until now our DEV@Deakin web platform was a single page. Now, we have multiple pages to navigate between. Your existing navigation bar and footer should appear identically across *all* of your pages.

Hint: modify your navigation bar and footer into their own React components, which you can easily add to any additional webpages you create. Even better is to add a Header and Footer component to your app's entry point (usually *main.tsx*).

Optional Technologies

We strongly encourage you to explore additional technologies that are often standard in industry, that may assist you in completing this task:

- [Zod](#) – An excellent library for data validation, such as usernames, emails and passwords. Robust data validation skills are crucial in industry.
- [TypeScript](#) – Adds 'type safety' to JavaScript. Standard JavaScript is not type-safe, meaning you can store any datatype in any variable which often leads to bugs. Strongly recommended to explore, either within this task or in your own time, and is expected to use within industry.
- [Vite](#) – A modern, faster alternative to the now out-dated Create React App. It is a lightweight frontend framework, usable with React.
- [NextJS](#) – A more powerful, but a more structured framework that combines both frontend and backend into the one package. A slightly steeper learning curve, but enables much more control over page rendering.
- Authentication Providers – Sophisticated third-party options exist to handle user authentication and authorisation, that are standard to use in industry. Can explore options such as [Clerk](#), [Firebase](#), or others.

Optional VS Code Extensions

VS Code has a gigantic library of extensions that can improve and speedup the development process. Here are some that may assist you:

- Tailwind CSS IntelliSense by Tailwind Labs – If using Tailwind CSS, this is a must-have extension to enable auto-complete when typing.
- Prettier by prettier.io – Automatically formats your codes' style and structure, to enforce a consistent style throughout.

Available Resources

Caution: Acceptable use of Artificial Intelligence

It is acceptable to use AI as a learning tool. However, you can **never** ask AI to write code to complete some/parts of this task for you: this is cheating because it does not demonstrate that you understand anything. Detection of inappropriate use of AI has a very real consequences, such as failing the task, or even failing the entire unit.

We have many resources to assist in your learning journey – if you are ever unsure then please reach out to us, we are always happy to help.

- **The SIT313 Teams channel** – an active learning resource with weekly announcements from the Unit Chair, and regular discussions/assistance from the SIT313 teaching team and students.
- **Seminars** – you can attend both on-campus and online sessions, and join as many as you like each week. Your tutors will provide demonstrations and assistance to help guide your learning.
- **SIT313 CloudDeakin** – contains a large collection of text and video resources.
- **Announcements** - Keep an eye on your email and any announcements made in the SIT313 Teams channel.

